



10 CFR 50.73

CCN: 23-97

December 14, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 3
Subsequent Renewed Facility Operating License No. DPR-56
NRC Docket No. 50-278

Subject: Licensee Event Report (LER) 2023-002-00 Main Steam Isolation Valves
Stroke Times Exceed Technical Specification Limit

The subject report is being submitted in accordance with 50.73(a)(2)(i)(B) for any operation or condition which was prohibited by the plant's Technical Specifications.

There are no commitments contained in this letter. If you have any questions, please contact the Peach Bottom Regulatory Assurance Manager, Mr. Wade Scott at (717) 456-3047.

Respectfully,

Stiltner, Ryan C

Digitally signed by Stiltner, Ryan

Date: 2023.12.14 12:40:36 -05'00'

Ryan C. Stiltner
Plant Manager
Peach Bottom Atomic Power Station

Enclosure

cc: USNRC, Administrator, Region I
USNRC, Senior Resident Inspector
W. DeHaas, Commonwealth of Pennsylvania
S. Seaman, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Peach Bottom Atomic Power Station, Unit 3

☒ 050
☐ 052

2. Docket Number

00278

3. Page

1 OF 3

4. Title

Main Steam Isolation Valves Stroke Times Exceed Technical Specification Limit

5. Event Date

Month	Day	Year
10	16	23

6. LER Number

Year	Sequential Number	Revision No.
2023	- 002 -	00

7. Report Date

Month	Day	Year
12	14	23

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
Facility Name	☐ 052	Docket Number

9. Operating Mode

3

10. Power Level

0

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Wade Scott, Regulatory Assurance Manager

Phone Number (Include area code)

(717) 456-3047

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	SB	ISV	A585	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 16, 2023, during performance of a surveillance test while beginning a refueling outage, the stroke times of Main Steam Isolation Valves (MSIVs) exceeded the Technical Specifications limit of five (5) seconds. The apparent causes attributed to two (2) of the MSIVs had the potential to exist during the previous operating cycle. Therefore, the two (2) MSIVs are being reported under 10CFR50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications because the apparent causes had the potential to have existed longer than allowed by Technical Specifications.

The condition was corrected and the valves were returned to satisfactory performance by adjusting the dashpots and exhaust restrictors as needed. There were no safety consequences associated with this condition, as the valve stroke times did not exceed the times considered in the safety analysis.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Peach Bottom Atomic Power Station, Unit 3	☒ 050	2. DOCKET NUMBER 00278	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 002	REV NO. - 00

NARRATIVEEvent Description

On October 16, 2023, during performance of a surveillance test, the stroke time of the 'A' Outboard Main Steam Isolation Valve (MSIV) exceeded the Technical Specifications limit of five (5) seconds. The valve stroke time was 6.2 seconds. This test was performed in Mode 3 (hot shutdown) while the unit was being shut down for a refueling outage. Transition to Mode 4 (cold shutdown) was made shortly after completion of the test, and the mode of applicability was exited. The cause of the slow stroke time was attributed to drift in the valve dashpot. Follow-up testing was performed during the outage and adjustments were made as needed. The stroke time was verified to be satisfactory before the unit was restarted from the refueling outage.

Additionally, the 'B' Inboard MSIV also exhibited a slow stroke time of 5.1 seconds. The cause of this condition was attributed to delayed actuation, likely due to debris or contamination in the solenoid valve. The 'B' Outboard MSIV stroked in 5.5 seconds, however this could not be replicated in subsequent testing. The stroke times for these valves were adjusted accordingly and were verified to be satisfactory before the unit was restarted from the refueling outage. The remaining valves stroked within specified Technical Specifications limits.

There are two (2) MSIVs welded in a horizontal run of each of the four (4) main steam lines, with one (1) valve as close as possible to the primary containment barrier inside, and the other just outside the barrier. The valves, when closed, form part of the nuclear system process barrier. The MSIVs are operated by pneumatic pressure and by the action of compressed springs. The upper end of the valve stem is attached to a combination air cylinder and hydraulic dashpot that are used for opening and closing the valve and for speed control, respectively. Speed is adjusted by a valve in the hydraulic return line alongside the dashpot.

Verifying that the isolation times of each MSIV is within the specified Technical Specifications limits is required to demonstrate operability. The isolation time test ensures that the MSIVs will isolate in a time period that does not exceed the times assumed in the design basis accident analyses. In all cases, the times in the accident analysis were not exceeded because the analysis is performed assuming at least 10 seconds for the MSIV closure time.

This report is being made under 10CFR50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications. Although there was no evidence of degradation during the prior refueling outage or operating cycle, the condition associated with the 'A' Outboard MSIV likely existed for longer than allowed by Technical Specifications. Similarly, there was no evidence of degradation for the MSIVs on the 'B' main steam line. The time frame associated with the condition which caused the 'B' Inboard MSIV to exhibit a slow stroke time cannot be determined, but the valve is conservatively being included in this report. The 'B' Outboard MSIV stroke time that was not repeatable has no firm evidence of existing prior to the time of discovery and is not included within the reportable scope of this LER. The performance of the valves has been entered into the Corrective Action Process for further evaluation of the test performance and evaluation methodology.

There were no other systems, structures or components that were inoperable that contributed to the event.

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CONTINUATION SHEET**

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1. FACILITY NAME☒ **050**

Peach Bottom Atomic Power Station, Unit 3

☐ **052****2. DOCKET NUMBER**

00278

3. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2023	- 002	- 00

NARRATIVE

The Energy Industry Identification System component function identifier and name of each component or system:

EIIIS: Main Steam, SB

AO-3-01A-086A, A Main Steam Line Outboard Isolation Valve

AO-3-01A-080B, B Main Steam Line Inboard Isolation Valve

Manufacturer: Atwood and Morrill

Model: 20851-H

Safety Consequences

The Technical Specification Surveillance Requirement (SR) 3.6.1.3.9 requires the MSIVs to close within 3 to 5 seconds. The Updated Final Safety Analysis Report (UFSAR) conservatively utilizes a maximum valve closure time of 10 seconds for analyses where the loss of reactor coolant inventory is the controlling variable, which provides conservatism between the SR and the analyzed performance. The observed stroke times for the noted MSIVs were no greater than 6.2 seconds. Therefore, the stroke times observed were not outside the bounds of the analyses and the safety function was maintained. There were no actual or potential safety consequences as a result of this event because this issue was identified during testing and the MSIVs closed within their analyzed times.

Since the remaining MSIVs closed satisfactorily, the overall containment isolation function would have been accomplished in 5.1 seconds.

Corrective Actions

This condition was corrected by performing follow up stroke timing for each MSIV and making adjustments such as dashpot and exhaust restrictor tuning as required to return the valves to desired performance.

The valve performance has been entered into the Corrective Action Program for further evaluation of the test performance and evaluation methodology.

Previous Similar Events

Peach Bottom Unit 3 LER 3-2009-007: An MSIV stroked below the minimum allowable stroke time due to dashpot oil leakage.

Peach Bottom Unit 3 LER 3-1998-003: An MSIV stroked outside the acceptance range.

Peach Bottom Unit 2 LER 2-1983-025: An MSIV stroked above the maximum allowable stroke time due to insufficient flow through the dashpot needle valve.